

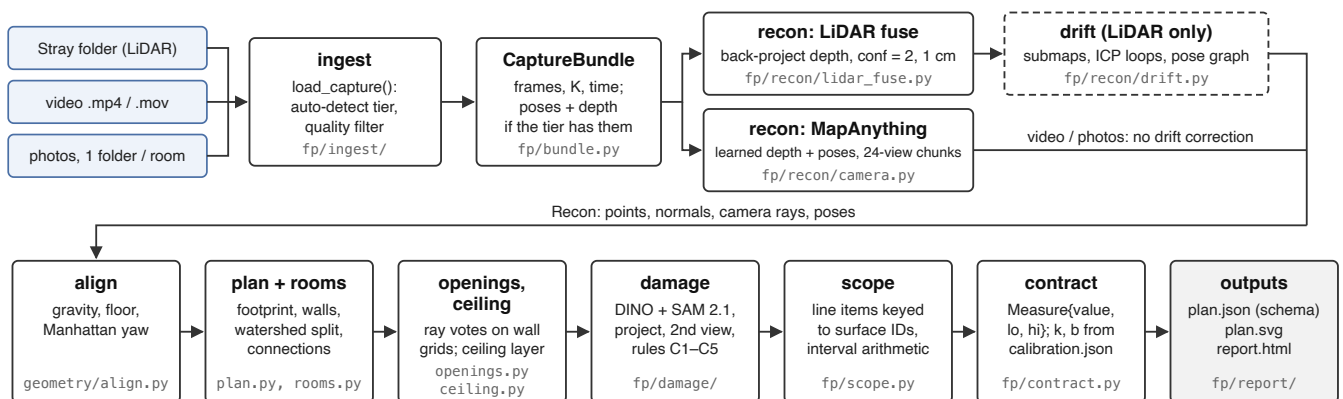
Floorplan: technical report

Phone capture → dimensioned, stitched floor plan. Repo `floorplan`, 8 Oct 2026. Every number links to the file that holds it (mostly `eval/`, regenerated by `make benchmark`).

1. Summary

- **What runs.** One command, `uv run fp run <capture>`, takes a Stray Scanner LiDAR folder, a video, or per-room photo folders. It writes one schema-valid `plan.json` with a 90 % interval on every number, plus `plan.svg` and `report.html`. All of it runs on a laptop (CPU/MPS). The benchmark ran all 7 sample plans in 962 s.
- **What passes.** 3 of 27 gates: the drift ablation on both whole-flat scans, and video interval calibration. That last pass comes from wider intervals, not better accuracy (§6).
- **What fails.** 13 gates. LiDAR repeatability is about ± 5 cm, not 1 cm. Openings are 50 % right by eye. Camera-tier walls are 37–39 % off at the median, and the photo stitch loses 70–79 % of the floor.
- **What isn't scored.** 10 gates are NOT SCORED and 1 is NOT DONE. The evaluator's sample data has no tape, and my own captures (video, photos, tape, app export) are still pending. Until then the stand-in for truth is the LiDAR plan, which is a reference and not ground truth.

2. Architecture



One command: `uv run fp run <capture>`. Dashed box = LiDAR tier only. A stage that fails or is missing writes a warning and observed: false; the plan is still schema-valid.

- **Ingest** (`fp/ingest/`). `load_capture` detects the tier from the folder. Stray (it has `depth/` and `odometry.csv`) is checked before photos, so depth maps are never read as photos. Poses come from `odometry.csv`, using the OpenCV camera convention and odometry timestamps (never `i/60`). Photos: HEIC, EXIF rotation, P3 → sRGB, and a focal-length prior from EXIF. Video: bundled ffmpeg, HDR tone-mapping, the rotation tag, 3 fps, at most 240 keyframes. A quality filter drops blurred frames, fast turns ($> 40^\circ/\text{s}$) and duplicates.
- **Recon, LiDAR** (`fp/recon/lidar_fuse.py`). Depth is back-projected with the ARKit poses, keeping only confidence 2, depth within 4 m, and a 1 cm voxel grid. I chose these settings by measured wall thickness (D8).
- **Recon, camera tiers** (`fp/recon/camera.py`). MapAnything runs locally and predicts metric depth, intrinsics and poses. Video runs in 24-view chunks with 6 views of overlap. Chunks are merged by the median depth ratio at shared pixels, and the scale is the median of the chunks' votes. Photos go through one joint pass, so all rooms share one frame. Scale priors (camera height, ceiling height) widen the intervals but never rescale the plan. Raw model outputs are cached by input-byte hash, so a replay gives an identical plan and a new capture always runs live.
- **Drift** (`fp/recon/drift.py`, LiDAR only). Pose graph over submaps (§4).
- **Align** (`fp/geometry/align.py`). Gravity comes from the sensor, or from image-up on camera tiers. The floor is the largest up-facing layer below the camera. Yaw is the peak of the wall-normal angle histogram (mod 90°).
- **Plan and rooms** (`fp/geometry/plan.py`, `rooms.py`). The footprint is the union of horizontal surfaces, cells crossed by camera→point rays from ≥ 3 frames, and a 0.3 m band around the camera path. Tall wall

evidence is cut out, and 0.5–1.3 m gaps on a wall line are barred as doorways. A watershed then splits rooms. Each room becomes a rectilinear polygon whose edges sit on fitted wall planes.

- **Openings and ceiling** (`openings.py`, `ceiling.py`). On each wall elevation, depth rays vote "wall", "seen through" or "unobserved". Doors, windows and passages are classified by sill and lintel height, and width is measured between the jamb planes. A mirror test and a recess test reject false openings. The ceiling is the largest down-facing layer 2.1–3.6 m above each room's own floor. If none is found, it is reported as not observed.
- **Damage** (`fp/damage/`). Grounding DINO plus SAM 2.1 find candidates. Each mask is projected onto its wall, floor or ceiling plane for a metric extent. A detection counts only if a second view ≥ 25 cm away confirms it. Rules C1–C5 raise concealed-damage flags, labelled as hypotheses.
- **Scope** (`fp/scope.py`) turns each surface into line items (repaint, patch, skirting), using interval arithmetic. **Contract** (`fp/contract.py`) turns every value into `Measure{value, lo, hi, unit, method, observed}`, with the fitted constants from `fp/calibration.json`. Damage, openings and scope refer to surfaces by stable IDs (`R1.W2`), and the schema rejects any reference that doesn't exist.

3. Tier design and device matrix

All three tiers use one schema and one CLI. Only tier and the interval width change. Before calibration, the evidence term is scaled $\times 1 / \times 3 / \times 6$ for LiDAR / video / photos (`fp/contract.py`). The fitted constants then widen it further ([CALIBRATION](#)). The full matrix is in [DEVICE_MATRIX.md](#).

tier	capture	phone	depth + poses from	laptop run (M5, MPS)	wall-length error today	stated interval
LiDAR	Stray Scanner walk	Pro iPhone / iPad with LiDAR	sensor + ARKit, drift-corrected	c7d2 flat 233 s	repeat spans median 5.5 cm	k = 1.42 $\times$ evidence
video	handheld walkthrough	any iPhone 15+	MapAnything (learned)	c7d2 511 s live	median 37–39 % vs LiDAR	$\pm$ 409 % of a wall
photos	2–8 per room, one folder per room	any iPhone 15+	MapAnything, one joint pass	17 photos, one pass	median 15–22 % vs LiDAR	$\pm$ 312 % of a wall

The camera-tier inputs are derived from the Stray captures: the video and stills are cut from `rgb.mp4`, with the rotation tag and EXIF focal length a phone would write (`scripts/make_camera_tiers.py`). The protocol tells users that video is the stronger camera tier.

4. Drift handling

Method (D12). ARKit's gravity and frame-to-frame motion are good. Heading and position drift over a multi-minute walk. I cut the walk into rigid submaps (every 3 m or 8 s). Where the camera revisits a place, I align the two submaps with point-to-plane ICP. A loop is accepted only if the fit is $\geq 20 \%$, the RMSE is ≤ 1.5 cm, the geometry pins all three directions (a bare corridor can slide, so it is rejected), the correction is plausible, and it tilts $\leq 1^\circ$. An Open3D pose graph with a line process then prunes inconsistent loops and solves for one correction per submap. I apply yaw and translation only, because gravity stays ARKit's. Corrections are blended in time between submap centres. The flag `--no-drift-correction` turns it off for the ablation.

drift correction OFF (ARKit poses as-is)
 median wall thickness 3.62 cm
 double walls (red): 32
 8 rooms, footprint 63.4 m²

drift correction ON
 median wall thickness 3.27 cm
 double walls (red): 15
 9 rooms, footprint 62.6 m²
 loop RMSE 3.22 → 2.12 cm



Figure: c7d28f72c6, drift off (left) and on (right). Red marks double walls. Source: out/c7d28f72c6/debug/drift_ablation.png; the same figure for 1a8384c3f6 is in docs/img/drift_ablation_1a83.png.

capture	wall thickness off → on	double walls off → on	loop RMSE off → on	loops accepted	footprint off → on
1a8384c3f6 (115 s, no ceiling)	4.2 → 1.9 cm	13 → 7	3.34 → 1.83 cm	8 / 25	55.31 → 57.05 m ²
c7d28f72c6 (215 s, revisits)	3.6 → 3.3 cm	32 → 15	3.22 → 2.12 cm	17 / 138	63.43 → 62.64 m ²

(The double-wall counts are printed on the ablation figures, from plan.drift.metrics. They are not in eval/.)

What it costs, and what it can't fix. With drift correction **off**, the two scans' wall-to-wall spans agree better (median 1.9 cm vs 5.5 cm). But fewer walls register (55 % vs 69 %) and room edges agree 3× worse (40 cm vs 12 cm). Each correction is only good to about the residual loop RMSE (~2 cm), and that error ends up inside the rooms. So correction trades a few cm of local accuracy for global consistency. Drift larger than the plausibility test allows is rejected, not corrected. In a verifier test, 10° of yaw injected into half of 1a83 was mostly left in place. The plan now warns when correction widens the heading spread across submaps by more than 0.5°. The camera tiers get no drift correction at all, which is the root of §7.

5. Error budget

Wall lengths, areas and ceilings use the interval model from §6. The table lists the main error sources I can measure, per tier. "vs LiDAR" means against the LiDAR plan of the same capture, not tape.

measurement	LiDAR	video	photos
wall length	Plane fit ~1 cm + wall smear (median thickness after drift 1.9–3.3 cm). Drift-correction residual: repeat spans median 5.5 cm, p90 11.8 cm. Room split at a different doorway: per-wall edges median 12.5 cm.	Camera poses: video path vs LiDAR Sim(3) scale 0.48, residual 2.4 m; chunk merges disagree by 0.3–1.5 m. Result: median 37–39 %, p90 71–76 %.	Too few photos overlap. Median 15–22 %, p90 24–62 % on 3–9 matched walls.
floor area	Segmentation dominates: matched rooms differ by 0.5–2.6 m ² between scans, while their wall planes sit a median 3.8 cm apart.	Rooms merge (no doorway necks). Footprint +19 % / +22 %.	Rays from ≥ 3 frames are rare with few photos, so rooms come out undersized: –70 % / –79 %.
ceiling	Layer spread / √patches + 1 cm, + 3 cm if the room's floor wasn't seen. Never compared with tape. Not observed on 1a83 and c00a, as required.	One room scored: 2.74 vs 2.47 m (+11 %).	One room scored: 2.47 vs 2.45 m (+0.9 %).

measurement	LiDAR	video	photos
opening width	Jamb $\sigma + 1$ cm per jamb (+2 cm when a wall end stands in for a jamb). Widths not taped. Detection is the bigger error: 7/14 right by eye .	Not evaluated.	Not evaluated (too few frames per cell).
damage extent	Mask coverage $\times$ pixel footprint on the plane. Interval 20 % + the tier's scale term. Painted stain +11 % .	Same method on model depth; not evaluated.	Same; not evaluated.

The ceiling, opening and damage intervals are provisional, because there is no evidence to fit them ([CALIBRATION](#)). On c7d2 LiDAR the ceiling half-widths are ± 1.2 –4.2 cm and the door half-widths ± 5 –12 cm (out/c7d28f72c6/plan.json; not in eval/).

6. Calibration analysis

Model (fp/calibration.py, D07.1). Every Measure uses $\text{half} = k \cdot a + b \cdot \text{value}$. Here a is the measurement's own evidence: the plane fit plus the measured wall smear, $\times 3$ for a wall never seen. $b \cdot \text{value}$ is a relative (scale) term. LiDAR fits k with $b = 0$, because its scale is metric. The camera tiers fit b with $k = 1$, because their error is scale and shape. The fit is split conformal: each evidence row gives the factor its interval would have needed, and the constant is the $[(n+1) \cdot 0.9]$ -th smallest. If $n < 9$, I take the largest (flagged). Without tape, a constant may widen an interval but never narrow it.

Evidence and held-out coverage ([CALIBRATION](#)). Each fit is checked 2-fold: fit on one half, then count how often the other half's reference falls inside the interval.

tier	evidence (stand-in for tape)	fitted	held-out coverage of stated 90 %	gate
LiDAR	16 wall-to-wall spans, repeat scans; whole disagreement charged to each scan	$k = 1.42$	62 % / 100 % $\rightarrow$ 81 % of 16	FAIL
video	12 walls vs the LiDAR plan, folds by capture	$b = 4.09$	100 % / 86 % $\rightarrow$ 92 % of 12	PASS
photos	12 walls vs the LiDAR plan, folds by capture	$b = 3.12$	100 % / 56 % $\rightarrow$ 67 % of 12	FAIL

Where it is optimistic.

- LiDAR spans include only wall planes the two scans matched within 10 cm, so a larger disagreement never enters the fit.
- Repeatability shows precision only. A bias shared by both scans is invisible without tape.
- On per-wall polygon edges, the LiDAR stress set, coverage is only [61 % of 23](#). The intervals model plane error, not a room split at a different doorway.
- Camera constants are fitted on the same two captures the gates score, so in-sample coverage is 100 % by construction.
- Floor areas have $n = 4$ –6, so the "largest of n " rule applies.
- Ceilings, openings and damage are not calibrated.
- n is tiny: one miss moves coverage by 6–8 points.

Why the video pass is not progress. The fix loop matched more video walls. The refit then widened the video wall constant from $b = 2.41$ to [4.09](#), so an interval is now ± 409 % of a wall. Held-out coverage rose from [78 % of 9](#) to [92 % of 12](#) because the intervals say "we don't know" more loudly, not because the walls got better. This is the honest outcome the brief asks for: wide intervals on thin input, rather than confident garbage.

7. Fix loop

Worst gate (declared before the fix in [FIX_DECLARATION](#)): video wall lengths within ± 3 % on c7d28f72c6–video, p90 [105 %](#) (35 $\times$ the threshold), with the footprint at 13.2 vs 62.64 m².

Root cause. I replayed extract_rooms one step at a time. The footprint was fine (84 m²). The area was lost in the room split: [78 % of the free floor](#) fell into seedless slivers between "tall walls". [82 %](#) of those tall cells

were seen through by ≥ 3 frames, so they were misregistered copies of walls standing in air. The whole-flat wall cut (from §2) assumes a drift-corrected cloud, and the camera tiers have none.

Fix (fp/geometry/plan.py, seen_through_walls). On predicted-depth tiers only, a tall cell that rays from ≥ 3 frames cross is treated as free space. It reuses the existing MIN_RAY_HITS rule, so no new threshold was tuned. The LiDAR tier is unchanged by construction, so its reference plans stay identical. Rejected alternatives, each tested by changing one thing: the rule on every tier removed 42–75 % of the real LiDAR walls, and giving seedless floor to the nearest room made p90 142 %.

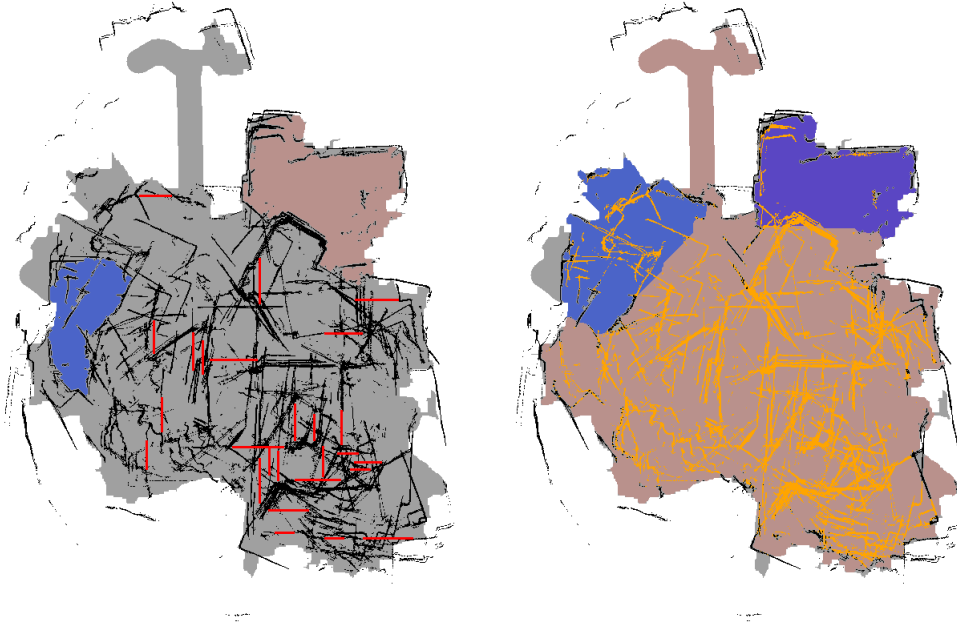


Figure: fixloop/before|after/c7d28f72c6-video/rooms_split.png. Before (left): a lattice of black "walls" shreds the floor, and only two small rooms are kept. After (right): the lattice is ignored (orange), and the floor forms three rooms.

number (RESULT)	before	predicted	after
c7d2 video wall p90 err (the gate)	105 %	76 %	76 % (FAIL, 25.5×)
c7d2 video wall median err	75 %	37 %	37 %
c7d2 video footprint	13.2 m ² (−79 %)	+22 %	76.47 m ² (+22 %), IoU 0.18 → 0.59
c00a video footprint / wall p90	−12 % / 63 %	+19 % / 71 %	+19 % / 71 % (worse)
LiDAR and photo plans	—	unchanged	unchanged

Why it fell short. The fix stops the plan stage from throwing floor away. It cannot repair the geometry it is given. The video camera path is metres off: Sim(3) scale **0.48**, median residual **2.4 m**. Passing ± 3 % needs better camera-tier poses, for example a pose-graph and loop-closure pass like the LiDAR tier's. That means new model passes, which is out of scope for a plan-stage fix. The fix also has a cost: on camera tiers, a real wall that frames see through (glass, or a misregistered frame) is removed, and c7d2 video R1 is still a **61 m²** merged room. **Regenerate:** make fix-before / make fix-after. Each checks out its tag in a temporary worktree and reruns the full benchmark (~16 min).

8. Known failure modes

- **Mirrors.** The footprint needs rays from ≥ 3 frames, so one reflection can't punch through a wall. For openings, a mirror test reflects the points behind a gap back across the wall plane. If ≥ 50 % land on the room's own surfaces, it is a mirror. An oblique vanity mirror scores only 0.30 and survives as a **phantom window** (R5.O1). I did not lower the threshold, because real doors score up to 0.38. On camera tiers the model's depth may put a "room" behind a mirror. That is untested.
- **Glass.** LiDAR sees through glazing, so the glazed balcony slider is called a passage (R4.O1, wrong kind). On camera tiers, the fix-loop rule removes any wall that ≥ 3 frames see through, so a glass partition merges two rooms. A synthetic test confirmed this (RESULT).
- **Wet-look and shiny floors.** Specular floors return little or wrong depth. The confidence-2 filter drops unreliable pixels, so the floor goes missing rather than turning into a phantom. The footprint still covers it

through the camera-path band and rays to the furniture. A reflection that passes the filter could add points below the floor. Not tested: the sample flats have no wet floors.

- **Low light.** The quality filter drops blurred frames (sharpness < 30 % of the median) and fast turns. LiDAR depth is active, so it should hold up, but ARKit tracking and RGB (damage, openings sheets) degrade. Learned depth on dark, noisy frames will be worse. Not tested.
- **Open-plan spaces.** The watershed splits only at a neck, so a room with no doorway or wall at wall-band height stays merged. Part of the corridor merges into the open kitchen on both flat scans (STATUS 03). On camera tiers, blurred doorways give one big room.
- **Non-Manhattan walls.** `align` picks one dominant yaw, and `plan.py` finds wall lines only along x and y. A diagonal or curved wall produces no wall line. Its stretch of the polygon becomes a staircase of *inferred* (dashed) edges, whose evidence term is $\times 3$. The plan does not fit a diagonal. Not tested: the sample flats are rectilinear.
- **Openings missed.** Two curtained windows (no see-through rays), a 0.15 m-sill window (no rule below 0.3 m), and a closed door seen only above 1.35 m: [4 misses](#).
- **Damage.** A drawn 0.30 m crack was [missed](#). Static shadows and grout lines pass the second-view check, because they stay put too. On the sample data, every confirmed item was false or doubtful ([STATUS 06](#)).

9. Not done

- **Head-to-head** against a consumer app: [NOT DONE](#). There is no LiDAR iPhone, and the recruiter approved skipping own LiDAR captures. A substitute protocol (camera tiers vs magicplan) is ready.
- **Tape and own captures.** Wall, ceiling and opening widths vs tape, video and photo repeatability, and staged damage are all NOT SCORED until `data/own/` and `eval/ground_truth/own.yaml` exist. `make benchmark` picks them up automatically and refits the intervals.